HOW THE BIG U WILL PROTECT THE BIG APPLE

New York City bets on a sprawling system of berms, levees, floodwalls, and floodgates to protect its most flood-prone areas from another superstorm.

Written by Lina Zeldovich

ON OCTOBER 29, 2012, SUPERSTORM SANDY HIT NEW YORK CITY, causing unprecedented damage. Over the course of 48 hours, the hurricane destroyed about 300 homes and damaged 69,000 dwellings. When a Con Edison power plant flooded, one of its transformers blew up, with a series of light pulses reminiscent of an old-school apocalyptic movie, leaving hundreds of thousands of New Yorkers without power for days. Thousands were temporarily displaced and 44 died.

Overall, Sandy caused more than $70 billion in damages and lost economic activity in the United States—about $19 billion to New York City alone. It was a wake-up call for New York City with its 520 miles worth of waterfront, which is longer than Miami, Boston, Los Angeles, and San Francisco combined.

At first, Sandy was deemed a one-in-700-year storm, as it was a unique phenomenon on many levels. First, a region of high pressure pushed it away from the more typical path over the western North Atlantic. Second, it hit near perpendicular to the coast, rushing water inland with unprecedented force.

However, later studies have warned that a similar storm could be seen within a century. Sea level rise was another concern. To avoid becoming awash in the ocean, the city would need to protect itself against another record-breaking and deadly storm surge. Within a year, the U.S. Department of Housing and Urban Development launched a Rebuild by Design competition, asking the nation's leading minds to devise a plan to protect the lower part of Manhattan, which is most vulnerable to the storm surge flooding. The winner was Danish architecture firm Bjarke Ingels Group (BIG) with a proposed solution dubbed the Big U, which is—fittingly—a U-shaped flood protection system. Envisioned as a continuous 10-mile waterfront network of berms, levees, and floodgates, the original plan was for it to stretch down from Midtown West, curve around the lower tip of the island, and reach up to Midtown East.

A rendering that illustrates where the Big U will travel around Manhattan, bringing a mix of infrastructure, including flood walls and floodgates. Image: Rebuild by Design BIG Team

The project had to span multiple neighborhoods, each with their own topology, population, and land ownership, explained Simon David, who was a project lead for the East Side Coastal Resiliency effort at BIG at the time. David later launched his own company, Office of Strategy and Design (OSD), which is still involved in implementing Big U coastal resilience efforts.

"One of the central problems to solve when you're engineering a flood protection system is how do you maintain a functional city while trying to create the most efficient alignment protection line?" he said.

"One of the central problems to solve when you're engineering a flood protection system is how do you maintain a functional city while trying to create the most efficient alignment protection line

—Simon David, principal and creative director, Office of Strategy + Design (OSD?

The most efficient flood protection is achieved with one continuous uninterrupted wall, like a berm or a levee. "But the local economy and people's day-to-day activities require moving across streets," David added.

Residents also need easy access to highways, piers, and ferries, as well as recreational waterfront areas. That means continuous floodwalls must be interspersed by multiple floodgates that stay open daily, then slam shut when flooding is expected. There's even more complexity to consider given Manhattan neighborhoods' terrain and infrastructure vary greatly.

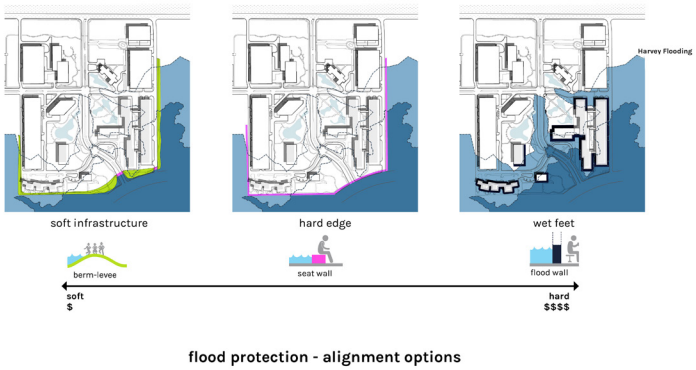
Manhattan's East Side already features ample park space, so there's enough land to build berms and levees. But it's also easier when there is only one landowner to deal with—the city, in this case. That changes drastically in the Financial District, where every building is owned by a different party, and some skyscrapers tower over the shoreline with no land upon which to build flood protection. Some Manhattan neighborhoods have historical and cultural landmarks to consider as well. These spots double as tourist attractions, bringing in significant economic revenue, and thus must still be accessible to the public. A one-size-fits-all approach simply wouldn't work.

As a result, the Big U was broken into four primary segments: the Brooklyn Bridge Montgomery Coastal Resilience (BMCR), Lower Manhattan Coastal Resilience (LMCR), Battery Park City Resilience (BPCR), and East Side Coastal Resilience (ESCR), which became the first step forward in implementing the Big U. Each segment faces its own challenges and unique designs, each of which is being handled by multiple engineering teams. As of 2025, the ESCR is the closest to achieving completion, while the others remain in design stages.

BREAKING GROUND

Multiple considerations came into play when it came time to decide where to begin work. The ESCR appeared to be the easiest portion, as it covers the longest stretch of uninterrupted terrain and primarily consists of parkland—all belonging to New York City.

"To come up with a unified system is much easier when it's all owned by the city, versus a bunch of privately owned buildings," David explained. "East Side Coastal Resilience was the ideal or the easiest solution to start with because you didn't have to worry about private individuals, businesses, and



Illustrating the differences among three primary flood protection alignment options. Image: Office of Strategy + Design

the existing infrastructure." Even so, the ESCR phase has proven to be far more complex than anticipated.

The initial design proposed transforming East River Park, built by Robert Moses in the 1930s, into terrain that could flood—essentially wetlands—which would slope upward to a floodwall that stretched around FDR Drive.

The concept of using nature as a buffer would have taken about four years to build, with a price tag of approximately $770 million. That idea was soon scrapped due to unforeseen challenges. “Maintaining a park that floods is extremely costly because it requires various cleanup and different types of vegetation,” said Matthijs Bouw, architect and founder of One Architecture & Urbanism, which is also involved in the Big U and the ESCR project. “So, one of the things that the Parks Department realized was that given their budgetary constraints and their priorities, this was not something that they could take on.”

tPlanners didn’t want to set a precedent either—the city has multiple parks within floodplains and the previous plans could have set unrealistic expectations that other flood-prone parks could be converted to wetlands. As a result, the ESCR project’s scope expanded in 2018 to include raising and redesigning additional portions of East River Park and included new NYC Department of Environmental Protection sewer system capacity and outfall upgrades.

Although the new plan was more expensive, doubling the price up to $1.45 billion, it was also more practical. The updated design still has a 4-foot to 10-foot floodwall, but now it spans the entire 1.2-mile length of the park, with the park itself sitting atop a raised landfill. Overall, the protective system will run from Montgomery Street north to East 25th Street.

“Maintaining a park that floods is extremely costly because it requires various cleanup and different types of vegetation. So, one of the things that the Parks Department realized was that given their budgetary constraints and their priorities, this was not something that they could take on.”

—Matthijs Bouw, architect and founder of One Architecture & Urbanism

This protective system includes two types of floodgates, a total of 18 overall. One type is a roller gate (seven of the 18), which resembles a sliding driveway gate and looks like massive moving pieces of metal.

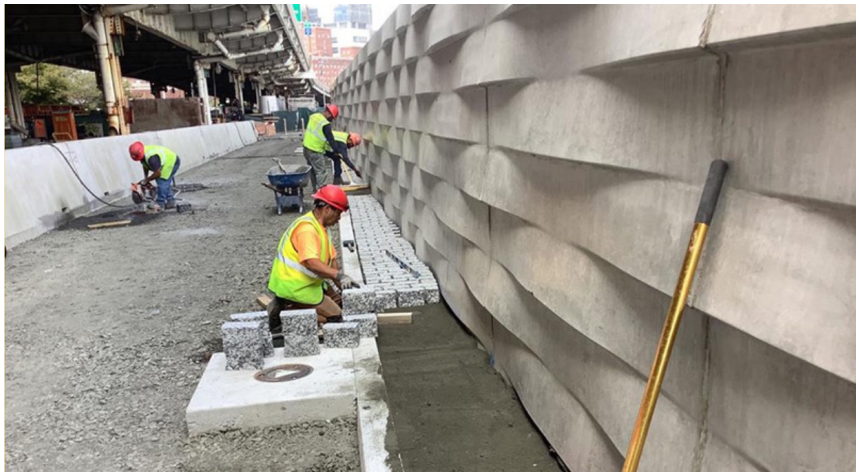
The first of these gates, measuring 40 feet long by 10 feet high and weighing 32,000 pounds, was installed in 2022. The largest one will be even longer, reaching 68 feet, said Ryan Stoddard, principal civil engineer at Arcadis, one of the engineering and design companies involved with the lineup of Big U projects.

These gates serve as openings in the otherwise uninterrupted floodwalls that allow urban life to continue unimpeded. They are typically used when a large space—such as a wide entrance to a park, for example—needs to be protected. “Those are there to connect urban spaces with each other,” unless a surge is expected, Bouw explained.

On the inside, the roller gates are built with steel frames, held together by huge front plates. They move on tracks with casters and are put in motion by a cable system that hooks to the gate at hook points and slowly pulls it along. “It moves at the scale of a train locomotive, almost,” Stoddard said.

The cable system relies on portable power and was deliberately built off the grid, so it can operate during power outages. And if all fails, the roller gates could theoretically be pulled closed by a vehicle.

The second type of gate—11 of the 18 total—is a smaller swing gate, typically



Crews recently installed a new floodwall alignment (above) and park (below) at East 23rd Street. These walls are typically supported by deep foundations, which were designed for scalability. The wall stem can also be raised 2 feet for future adaptability. Formliners provide an aesthetic finish, while epoxy coating reinforcement adds to the infrastructure's design life. Photos: Arcadis

used to provide access to smaller openings. Normally, they are tucked away on the sides of the highway, but when a surge is expected, they would swing to close the road like doors.

Swing gates are generally easier to operate and are more reliable because you don't have to rely on tracks and casters, Stoddard said. Not having to build tracks means you don't have to worry about tracks being damaged by vehicles or snowplows, which is why swing gates are better suited for highways or pedestrian walkways. But since swing gates operate on hinges, they have a length limitation. When the gate is stored, a jack holds up the end. Although these gates are designed to limit deflection when deployed, Stoddard added, swing gates are typically kept at about 40 feet long or less.

Despite various delays, including some legal issues, construction on the redesigned ESCR finally broke ground in fall 2020 and is progressing. In October 2024, crews completed the first big milestone: "We are marking the completion of Phase One of East Side Coastal Resiliency—two months ahead of schedule and $10 million under budget," said New York City Mayor Eric Adams in a statement. "Once the entire project is completed, the protective gates can be deployed when a hurricane or storm surge is headed our way—protecting lives and New Yorkers' wallets as we safeguard property."

It will be another two to three years before the ESCR reaches completion, Bouw said.

Immediately adjacent to the ESCR is the East Midtown Coastal Resiliency Program. A unit of its own and currently under design, it will run only five blocks, from 25th Street to 30th, along the East River, protecting the city's major hospitals, including NYU Langone and Bellevue.

THE REST OF THE U

Despite being considered the easiest to start with, the ESCR segment of the Big U has demonstrated some of the primary difficulties in storm-proofing New York City. Other portions of the effort will present even more challenges; For example, on the LMCR and BMCR, which would eventually fortify the Financial District, the Seaport, and other low-lying areas, initial logistics are a major hurdle.

"As you make your way down toward lower Manhattan, it's the buildings and the highway that are right up against the water instead of the park," David explained. "The negotiations that the flood protection system has to make

with the existing physical conditions are different—and in many ways much more complex because of existing infrastructure, existing foundations, and many private owners."
A possible solution could be building new land offshore. "Much like the land that was reclaimed for East River Park in the 1920s, it's possible that we might actually build new land into the East River," David said. "That would provide the space necessary for things like levies, thereby not even having to worry about all the private individuals, businesses, and the complex navigation of existing infrastructure."

However, in the case of the Financial District, its value as a landmark and a tourist attraction remains a consideration as the area is visited by millions every year, said Edgar Westerhof, vice president at Arcadis.

"When it comes to the Financial District, it's a recreational attraction with many, many tourists," Westerhof added. "So, essentially, on the back of a flood protection plan, you have to create value under normal circumstances."

The solution currently planned for the Financial District will span both land and sea. It will include land constructs such as walls and berms, but also

An engineer works inside the nose cone of a Skyroot launch vehicle, a part of India's growing space and tech ecosystem. Photo: Skyroot Aerospace

offshore elements such as lingering berms that reduce wave energy and can also help with water ecology. "What makes the Financial District unique is what we call a hybrid plan," Westerhof said. "That means we have part of the solution on land and part of the solution can happen in the East River.

The BMCR part of Big U—design for which was led by the New York City Economic Development Corporation—might also employ a different type of floodgates: flip-up gates or lift gates. These are usually built into the ground, where the highway meets the flood protection system to make it waterproof. "They're normally stored underground," explained Stoddard, noting that they can be put in motion by a hydraulic system. They could be problematic to install, however, if there is some existing infrastructure underground already. "If you have utility lines underground, you have to oftentimes relocate those utilities," Stoddard said, so this option works best when the ground below is

clear.

There's another gate type called a drop-down—similar to those used in the doorframes of Venice restaurants and homes, where a panel drops into the rubber gaskets on the sides of the door to form a protective barrier against the water. "You see them in Venice when there's 'Acqua Alta,'" said David, referring to Venice's "high water" phenomenon. Whether it will be used in any of the Big U segments is not yet clear.

By 2050, the Big U's full scope will be complete, but that won't mark the end of the flood-proofing efforts, experts say. A grim reminder of the city's shortcomings came in 2021, when Hurricane Ida dumped up to nine inches of rain per hour on some parts of the city—far exceeding New York's sewer capacity of 1.75 inches per hour. As a result, some residents drowned in the basements of buildings. One can envision a weather event in which the excessive amount of rain can coincide with extreme surge, doubling the pressure on low-lying areas, Westerhof noted. This means more solutions must be implemented in the future.

That may include increasing sewage capacity, which is already being done as part of Big U, as well as constructing rain gardens and sponge roofs that can temporarily hold excess rainfall, David said. His company is in fact working on such a roof that can hold 177,213 gallons of water for 24 hours down in Houston, where the solution is already alleviating stress on sewer drainage infrastructure.
Another weather event where excessive amount of rain coincides with extreme surge would double the pressure on low-lying areas, Westerhof noted, emphasizing that more solutions will need to be implemented in the future.

Whichever way the water is coming, the Big Apple will never be able to lower its guard. Luckily, it seems that as challenges pour in, New York is rising to meet them.

LINA ZELDOVICH is a science and technology writer based in Woodside, N.Y. Her most recent book, The Living Medicine: How a Lifesaving Cure Was Nearly Lost—and Why It Will Rescue Us When Antibiotics Fail, was published in October 2024.

Article originally published in Mechanical Engineering, Vol 147, No. 10, October 2025.
Copyright 2025 ASME. All rights reserved.